



AI with Thiru

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NVIDIA LocateAnything-3B

LocateAnything-3B is a free AI model from NVIDIA that **finds things in images**. You give it a picture and a sentence like “find the dog,” and it draws a box around exactly where the dog is. Most AI can only *describe* a picture — this one tells you the precise **location** as boxes or points.

It is small (3 billion parameters), fast, and works across many kinds of images — photos, screenshots, and documents — just by changing the sentence you give it.

What you can use it for

Object detection

Find people, cars, products — even objects it was never specifically trained on.

Find by description

Point to a specific thing from plain English, e.g. “the man in the red shirt.”

Screens & apps (GUI)

Locate buttons, icons and fields on screenshots — used for computer-use AI agents.

Text & documents

Detect text, tables and layout regions for OCR and document understanding.

Crowded scenes

Stays accurate in busy, cluttered images where older detectors fail.

Auto-labelling

Generate boxes and points automatically to build your own datasets.

Good to know. The model is **free for research and learning**, but NVIDIA's license does **not allow commercial use**. It also needs an **NVIDIA GPU** to run on your own machine — if you don't have one, use the online demo on Page 2.

Run it in Hugging Face Spaces

The easiest way to try the model — **no install, no GPU, no code**. NVIDIA hosts a free online demo right in your browser.

huggingface.co/spaces/nvidia/LocateAnything

1

Open the demo link

Go to the address above in any web browser. (A free Hugging Face account helps if the demo is busy.)

2

Upload your image

Click the image box and choose a photo, screenshot, or document from your computer.

3

Type what to find

In the text box, describe the object — for example `person` , `the red car` , or `the login button` .

4

Click Run

The demo draws boxes on your image showing exactly where each match is. Try different descriptions to see how it responds.

Best for: quickly testing what the model can do, demos for a video, or anyone without a powerful GPU. Nothing is installed on your computer.

Run it locally in VS Code

To use it in your own Python project, open **VS Code** and follow these steps. You'll need an **NVIDIA GPU** and Python installed.

1

Make a project folder

In VS Code: **File** > **Open Folder**, then open the terminal with **Ctrl + `**.

2

Install the libraries

Run this once in the terminal:

```
pip install transformers torch torchvision pillow
```

1

Create a file `run.py` and paste this

Put an image named `my_photo.jpg` in the same folder.

```
from transformers import pipeline

# Loads the model from Hugging Face the first time (needs an NVIDIA GPU)
pipe = pipeline(
    "image-text-to-text",
    model="nvidia/LocateAnything-3B",
    trust_remote_code=True,
)

messages = [{
    "role": "user",
    "content": [
        {"type": "image", "url": "my_photo.jpg"},
        {"type": "text", "text": "Locate all the instances that match: person."},
    ],
}]

result = pipe(text=messages)
print(result)  # -> labels + box coordinates
```

1

Run it

Press the ► Run button, or type `python run.py` in the terminal. It prints the labels and box coordinates.

Two tips. The first run downloads the model (~8 GB), so it takes a while. Box coordinates come back on a **0–1000** scale — multiply by your image's width and height to turn them into real pixels.

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